Personal Loan Default Risk Analysis

Business Case & Data Analytics Project Blueprint

1. Business Problem

A bank or lending institution receives thousands of personal-loan applications and needs to manage credit risk while continuing to serve creditworthy customers. Historical applicant and loan data can be used to understand which customer, financial, credit-behavior, and loan characteristics are associated with higher observed default rates.

Core business question: Which types of applicants are associated with higher observed default rates, and how can the lending team use these patterns to improve risk monitoring and portfolio decisions?

2. Main Business Objective

Identify the major factors associated with personal-loan default and develop a data-driven risk segmentation framework that can support lending and portfolio-monitoring decisions.

3. Key Business Questions

Overall Portfolio Risk

- What is the overall application default rate?
- How many applications defaulted versus did not default?
- What is the loan amount exposure associated with defaulting applicants?
- Is default concentrated in particular customer or loan segments?

Credit Risk

- How does observed default rate vary across CIBIL-score bands?
- Is higher credit utilization associated with higher observed default?
- How does late-payment history relate to default?
- How do existing loans and credit inquiries vary across risk segments?

Income & Affordability

- How does default rate vary by monthly income?
- Is higher existing EMI burden associated with higher observed default?
- How does requested loan amount relate to the applicant's financial capacity?

Loan Characteristics

- Which loan purposes have higher observed default rates?
- How does loan tenure relate to default?
- How do applicants with and without collateral differ in observed default patterns?
- Does interest rate vary across observed risk segments?

Customer & Geography

- How does observed default rate vary by employment type?
- Do city tiers or states show different portfolio risk patterns?

- Are there meaningful differences in application volume and loan exposure across segments?

Risk Score Evaluation

- Does the existing default-risk score separate applicants into groups with different observed default rates?
- How many applications and how much loan exposure fall into each risk-score band?

4. Risk Segmentation Framework

The analysis should create business-friendly segments rather than relying only on individual columns. Suggested segmentation dimensions include credit quality, affordability, payment behavior, and loan characteristics.

Dimension	Example Segments	Business Purpose
CIBIL Score	<650; 650–699; 700–749; 750+	Compare observed default across credit-quality bands
EMI Burden	Low; Moderate; High	Assess affordability pressure
Credit Utilization	Low; Moderate; High	Identify patterns in revolving-credit usage
Late Payments	0; 1–2; 3–5; 6+	Measure relationship with repayment behavior
Loan Purpose	Purpose-specific groups	Identify portfolio segments with different default patterns
Risk Score	Low; Medium; High; Very High	Evaluate separation of observed risk across score bands

5. Important Derived Metric

Existing EMI-to-Income Ratio = Existing EMI ÷ Monthly Income × 100

This metric can be used to understand affordability pressure. Applicants can be grouped into low, moderate, and high EMI-burden segments and their observed default rates compared.

6. Analytics Workflow

Stage	Work	Output
1. Data Preparation	Missing values, duplicates, data types, validation, outlier checks	Analysis-ready dataset
2. SQL Analysis	Portfolio KPIs, segmentation, default-rate analysis, exposure analysis	Business findings
3. Exploratory Analysis	Credit, income, affordability, loan and geographic patterns	Risk insights
4. Power BI	Executive dashboard + credit/financial risk + segmentation pages	Interactive dashboard
5. Recommendations	Translate observed patterns into monitoring and decision-support insights	Business recommendations

7. SQL Analysis Plan

- Calculate total applications, defaulted applications, total loan amount, and overall default rate.
- Compare default rate by CIBIL-score band, income band, loan amount band, and employment type.
- Analyze default rate by loan purpose, tenure, collateral, city tier, and state.
- Analyze late-payment history, credit utilization, credit inquiries, existing loans, and EMI burden.
- Compare loan amount exposure between defaulting and non-defaulting segments.
- Evaluate default-rate differences across default-risk-score bands.
- Identify combinations of risk indicators associated with higher observed default rates.

8. Power BI Dashboard Structure

Page	Purpose	Suggested KPIs / Visuals
1. Executive Risk Overview	Portfolio health at a glance	Applications, loan exposure, defaulted applications, default rate, average CIBIL; defa
2. Credit & Financial Risk	Understand risk drivers	CIBIL, utilization, late payments, EMI burden, income, existing loans, loan amount
3. Risk Segmentation	Monitor risk groups	Risk-score bands, CIBIL × EMI burden matrix, high-risk segment count, loan exposur

9. Final Business Outcome

The final project should explain what characteristics distinguish applicants with higher observed default rates, how much loan exposure is associated with these segments, and which indicators the lending team should monitor when evaluating and managing personal-loan risk.

10. Portfolio Project Statement

Personal Loan Default Risk Analysis — Banking & Lending Analytics
Analyzed historical personal-loan applications using SQL and Power BI to identify factors associated with loan default. Evaluated credit quality, affordability, repayment behavior, loan characteristics, and risk-score segments; developed a risk segmentation framework and interactive dashboard to support credit-risk monitoring and lending decision support.

11. Analytical Guardrails

This is a historical analytics project. Findings should be described as associations or observed patterns, not proof that a variable causes default. Demographic variables should be treated carefully and primarily for portfolio analysis and fairness monitoring rather than as direct grounds for discriminatory lending decisions. Any real-world lending policy would also require regulatory, compliance, model-validation, and fairness review.